Operating System Lab
1ICPC204

Experiment No. 2

Study and Demonstration of basics of Linux/UNIX commands

Course Instructor -
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Title of Experiment: Study and Demonstration of basics of Linux/UNIX command.

Aim of Experiment: To implement and understand the basic Linux/UNIX commands.

System Requirements – Operating System: Linux distribution (e.g., Ubuntu)
Terminal Emulator (usually pre-installed in Linux distributions)
Text Editor (e.g., Nano)

Software/s Needed for Experiment – A Linux/UNIX operating system (e.g., Ubuntu)
Terminal emulator (default in Linux systems)

Experiment Outcomes –

1. Ability to use basic Linux/UNIX commands for file and directory management
2. Understanding of file permissions and how to modify them
3. Capability to use text editors for file creation and editing
4. Knowledge of process management commands

Theory –

Linux is a Unix-Like operating system. All the Linux/Unix commands are run in the terminal provided by the Linux system. The Linux Command Line Interface (CLI) is a text-based interface that will enable us to type text commands to instruct the computer to do specific tasks. The Linux command line is case-sensitive. The terminal can be used to accomplish all administrative tasks. This includes package installation, file manipulation, and user management. Linux terminal is user-interactive. The terminal outputs the results of commands which are specified by the user itself. In this experiment, we will run different commands in Linux with syntax so that we can understand their functionality in a better way.
Key concepts include:

- **File System Structure:** Understanding the hierarchical structure of directories.
- **Basic Commands:** Commands like ls, cd, cp, mv, rm, and mkdir.
- **File Permissions:** Understanding and modifying permissions with chmod, chown, and chgrp.
- **Text Editors:** Using text editors like Vim, or Nano for editing files.
- **Process Management:** Commands like wait, top, kill, stop, and fork.
- **System Information:** Commands like id, uname.
- **Networking Commands:** Commands like ping and ifconfig.

Demonstration of Basic Linux Commands

1. Command : pwd

   Description : To find out the path of the current working directory (folder).

   Syntax :
```
a@DESKTOP-72M267R:~$ pwd
```
   Output :

2. Command : mkdir

   Description : To create/make a new directory with given name

   Syntax :
```
a@DESKTOP-72M267R:~$ mkdir LINUX_COMMANDS
```
   Output :

3. Command : ls

   Description : To view the contents of a directory

   Syntax :
```
a@DESKTOP-72M267R:~$ ls
```
   Output :

4. Command : cd

   Description : To navigate through the Linux files and directories. Change to directory.

   Syntax :
```
a@DESKTOP-72M267R:~$ cd LINUX_COMMANDS
```
   Output :

5. Command : touch

   Description : To create a blank/empty new file

   Syntax :
```
a@DESKTOP-72M267R:~$ cd LINUX_COMMANDS
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ touch file1.txt
```
   Output :

6. Command : cat > filename

   Description : Create new file and writes the content and Ctrl+D to save the content into file

   Syntax :
```
a@DESKTOP-72M267R:~$ cd LINUX_COMMANDS
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat>file2.txt
```
   Output :

7. Command : cat

   Description : To display content of file

   Syntax :
```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ ls
file1.txt  file2.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat>file2.txt
Hello to All! Welcome
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file2.txt
```

Output :

8. Command : cat filename1 filename2>filename3
Description : Creates new file(file3) and content of both files (file1 and file2) will be copied to new file.
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ touch file1.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat>file2.txt
Hello to All!
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat>file3.txt
Welcome to Basic Commands
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file2.txt file3.txt>file4.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ ls
file1.txt  file2.txt  file3.txt  file4.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file4.txt
```

Output :

9. Command : ls *.txt
Description : List all the files with given extension
Syntax :

```
a@DESKTOP-72M267R:~$ ls
Desktop  LINUX_COMMANDS  a.out  check.c  commands.txt  fcfs.c  file1  hello.c  hello.out  hello.txt  test.py
a@DESKTOP-72M267R:~$ ls *.txt
```

Output :

10. Command : cp
Description : Copy a file or directory
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file2.txt
Hello to All!
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file3.txt
Welcome to Basic Commands
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cp file2.txt file3.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file3.txt
```

Output :

11. Command : mv
Description : Moves a file or directory
Syntax :

```
a@DESKTOP-72M267R:~$ ls
Desktop  LINUX_COMMANDS  a.out  check.c  commands.txt  fcfs.c  file1  hello.out  hello.txt  madhuri  test.py
a@DESKTOP-72M267R:~$ pwd
/home/a
a@DESKTOP-72M267R:~$ mv hello.txt /home/a/LINUX_COMMANDS
a@DESKTOP-72M267R:~$ ls
Desktop  LINUX_COMMANDS  a.out  check.c  commands.txt  fcfs.c  file1  hello.out  madhuri  test.py
a@DESKTOP-72M267R:~$ cd LINUX_COMMANDS
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ ls
file1.txt  file2.txt  file3.txt  file4.txt  hello.c  hello.txt
```

Output :

12. Command : tac
Description : To display lines of file in reverse order
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ tac file4.txt
```

Output :

13. Command : id
Description : To display id of user/group
Syntax :

```
a@DESKTOP-72M267R:~$ id
uid=1000(a) gid=1000(a) groups=1000(a),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo)
lugdev),118(netdev)
```

Output :

14. Command : head
Description : To display first ten lines of the file
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cd
a@DESKTOP-72M267R:~$ cd LINUX_COMMANDS
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ ls
file1.txt  file2.txt  file3.txt  file4.txt  hello.c  hello.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ vi file4.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file4.txt
line2
line3
line4
line5
line6
line7
line8
line9
line10
line11
line12
line13
line15

a@DESKTOP-72M267R:~/LINUX_COMMANDS$ head file4.txt
```

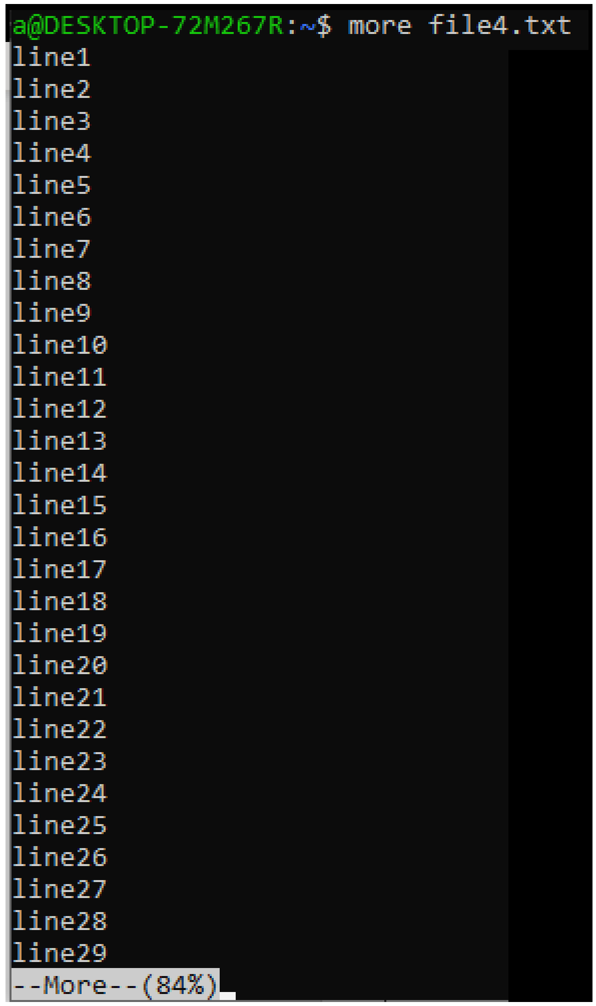
Output :

15. Command : tail
Description : To display last ten lines of the file
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ tail file4.txt
line12
line13
line15
line16
line17
line18
line19
line20
```

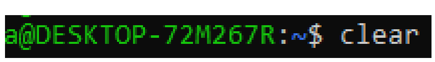
Output :

16. Command : more
Description : This command is similar to cat and here we can display large content by using ENTER or SPACEBAR.
Syntax :



Output :

17. Command : clear
Description : To clear the screen
Syntax :



Output :

18. Command : vi filename
Description : text editor to write the programs or text. Press :wq to save file and exit. If we want to exit text editor without saving the file press :q!
Syntax :
```
a@DESKTOP-72M267R:~$ vi file4.txt
```
Output :

19. Command : grep
Description : To search through all the text in a given file
Syntax :
```
a@DESKTOP-72M267R:~$ grep i file4.txt
```
Output :

20. Command : diff
Description : Compares the content of two different file. It removes the contents which is same and displays the content of file1 which is not available in file2
Syntax :
```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file2.txt
hello students
how are you all
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file3.txt
hello and welcome once again. how was your vacation.
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ diff file2.txt file3.txt
1,2c1
< hello students
< how are you all
---
> hello and welcome once again. how was your vacation.
```
Output :

21. Command : ping
Description : To check your connectivity status to a server. Press ctrl+Z to exit.
Syntax :
```
a@DESKTOP-72M267R:~$ ping google.com
```
Output :

22. Command : history
Description : To review the commands you have entered below.
Syntax :
```
a@DESKTOP-72M267R:~$ history
```
Output :

23. Command : hostname
Description : To display the hostname
Syntax :
```
a@DESKTOP-72M267R:~$ hostname
```
Output :

24. Command : hostname -i
Description : To display host ip
Syntax :
```
a@DESKTOP-72M267R:~$ hostname -i
```
Output :

25. Command : chmod
Description : To change the user/group permissions to access file. Press chmod +rwx filename to add permissions.
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file3.txt
Hello students!
Welcome to Basic Linux Commands



a@DESKTOP-72M267R:~/LINUX_COMMANDS$ chmod u=r file3.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ chmod u=w file3.txt
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ chmod +rwx file3.txt
```

Output :

26. Command : nl
Description : To display the line numbers
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ nl file3.txt
```

Output :

27. Command : wc
Description : To display number of lines, words and characters available in the file content.
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ wc file3.txt
```

Output :

28. Command : uniq
Description : Removes duplicates of file content/ it can remove only continuous duplicates
Syntax :

```
a@DESKTOP-72M267R:~/LINUX_COMMANDS$ cat file3.txt
Hello students!
Welcome to Basic Linux Commands
Welcome to Basic Linux Commands
Thank you
Lets Learn
Thank you



a@DESKTOP-72M267R:~/LINUX_COMMANDS$ uniq file3.txt
```

Output :

29. Command : rmdir
Description : Removes the specified directory(But the directory should be empty)
Syntax :

```
a@DESKTOP-72M267R:~$ mkdir abc
a@DESKTOP-72M267R:~$ ls
Desktop  LINUX_COMMANDS  a.out  abc  check.c  commands.txt  fcfs.c  file1
a@DESKTOP-72M267R:~$ rmdir abc
```

Output :

30. Command : rm
Description : Removes file and file need not be empty.
Syntax :

```
a@DESKTOP-72M267R:~$ rm file1.txt
```

Output :

Procedure:

1. Open Terminal: Start your terminal application from your desktop environment.
2. Check Current Directory: Type'pwd' to display the present working directory.
3. In this way we will run different commands in Linux with syntax on terminal so that we can understand their functionality in a better way.

Conclusion :

Hence, the experiment provides students with understanding and Hands-on experience of basic commands in Linux Operating system.

References –

a. Textbook –

i. The Linux Command Line: A Complete Introduction, William Shotts - 2nd Edition (2019), Publisher: No Starch Press
ii. Linux Pocket Guide - Daniel J. Barrett, 3rd Edition (2016) ,Publisher: O'Reilly Media

b. Online references –

i. https://github.com/labex-labs/linux-basic-commands-practice-online
ii. https://dev.to/labex/online-linux-playground-practice-linux-commands-online-5b25
iii. https://www.practicelinux.com

Expected Oral Questions –

1. What is an operating system?
2. Define Linux Kernel. Is it legal to edit Linux Kernel?
3. What are the basic components of Linux?
4. what is file permission in Linux?
5. What is the Linux Directory Commands?
6. What is the use of commands pwd?
7. What is the use of commands whoami?
8. What is a directory file?
9. What does the arguments in the "chmod category operation permission filenames" defines?
10. what is VI editor?